

Texas CDL Class A — Knowledge Test Study Guide

Key testable facts for the four written exams · Based on the Texas Commercial Driver License Handbook

How the written tests work

For a Class A CDL (tractor-trailer) in Texas you take four written knowledge tests. You must score **80%** on each. In Texas the **Special Requirements** test must be passed **first**, before any other CDL test.

Test	Questions	To pass	Covers
Special Requirements (TX)	~25	80%	Texas-only rules: lighting, size & weight, loads, equipment, breakdown — Section 14
General Knowledge	50	40	Inspection, control, speed & stopping, space, hazards, cargo, emergencies
Air Brakes	25	20	Air-brake parts, dual systems, inspection, using air brakes (removes the air-brake restriction)
Combination Vehicles	~20	16	Coupling/uncoupling, trailer air brakes, rollover, off-tracking (required for Class A)

Pass all four to earn your Commercial Learner's Permit (CLP); the road (skills) test comes afterward.

1 · Special Requirements (Texas — Section 14)

Equipment

- Hazard (flasher) lights: any truck, bus, tractor, or trailer **80 in. or more wide OR 30 ft or more long** must turn on hazard flashers when stopped on a roadway or shoulder — except when legally parked inside city limits or loading/unloading passengers.
- Fire extinguisher: school buses, buses, taxis and other for-hire passenger vehicles must carry a chemical extinguisher of **at least 1 quart capacity**; vehicles under federal rules must have one filled, securely mounted, and readily accessible.
- After-dark emergency gear (outside city limits): carry **at least 3 flares, or 3 red electric lanterns, or 3 portable red reflectors**; when lights aren't required, 2 red flags. Vehicles hauling explosives/flammables must **not** use flame-type signals — use DOT triangular reflectors.
- First action of a disabled-vehicle driver: **put out the flares, flags, or reflectors**.
- Reflectors mounted **24-60 inches** above the ground on pole trailers and on trucks/buses/tractors/trailers 80+ in. wide.
- Light/reflector colors: clearance & side-marker lamps and reflectors near the **front are AMBER**; those at the **rear are RED**. (Common Section 14 question.)
- Turn signals: all motor vehicles, trailers, semitrailers and pole trailers must have electrical turn signals — **except** trucks under 80 in. wide made **before model year 1960** (i.e. 1959 & earlier), unless the body/load extends >24 in. from the steering-wheel center or the rear of the load is >14 ft from it. Trucks built before model year 1960 must have **at least one taillight**.
- Mud flaps: all trucks and trailers with **4+ tires on the rear axle** must have flaps reaching **within 8 inches** of the road (not pole trailers or a bobtail tractor).
- Slow-Moving Vehicle (SMV) emblem: required on vehicles built for **25 mph or less**; illegal to display on any faster vehicle.

- Animal-drawn / farm equipment at night: white light on front visible **1,000 ft** + two red lights on rear visible 1,000 ft (or one red light + two red reflectors visible 600 ft).
- Farm tractors/implements built after **Jan 1, 1972**: two headlamps, one taillight (far left), two red reflectors, and hazard flashers (white/amber front, red/amber rear).
- Trailer brakes: $\leq 4,500$ lb gross are exempt; 4,500–15,000 lb at ≤ 30 mph not required; over 4,500–15,000 lb above 30 mph need brakes on both rear-axle wheels; one control must operate all brakes.
- Stopping ability: a combination $\leq 3,000$ lb must stop within **40 ft at 20 mph**; over 3,000 lb within **50 ft at 20 mph**.

Size, weight & load limits

- Height: max **14 ft** (ground to top of load). The driver is responsible for clearing every bridge/overpass.
- Width: max **102 inches** (8 ft 6 in), including the load.
- Lengths: single vehicle (not a truck tractor) **45 ft**; semitrailer in a tractor-semi-trailer ≤ 59 ft; each trailer in a tractor-semi-trailer-trailer $\leq 28\frac{1}{2}$ ft; any combination other than a truck-tractor-trailer ≤ 65 ft.
- Combinations: no more than **3 vehicles** in a combination; a vehicle under 2,500 lb unloaded may couple with only one other vehicle.
- Weight: greatest gross weight (vehicle + load) is **80,000 lb**; you may exceed it only with an oversize/overweight permit. A 5% tolerance applies before forced unloading.
- Extensions: a load may not extend **more than 3 ft beyond the front** nor **more than 4 ft beyond the rear** without a permit.
- Rear overhang marking: when a load extends more than 4 ft beyond the rear, attach a **red flag at least 12 in. square by day**, and a **burning red light visible 500 ft at night**.
- Towing: the drawbar/chain/rope/cable between two vehicles must not exceed **15 ft** (except pole trailers); if a chain/rope/cable is used, attach a white flag at least 12 in. square.
- Metal tires: a vehicle of 5,000 lb or more with metal tires needs a special permit. No cleats/lugs/spikes that damage the road (tire chains are OK).
- Loose materials (dirt, sand, gravel, wood chips): must be fully enclosed or covered to prevent blowing/spilling — except agricultural products in their natural state.

Operating rules

- Coasting: it is **unlawful** for a commercial vehicle to coast down a grade with the clutch disengaged or in neutral; shift to a lower gear **before** starting down the hill.
- Following: when one truck follows another, it must leave enough space for another vehicle to **overtake and move in between** them safely.
- Railroad crossings: buses carrying passengers for hire and school buses must **stop at all RR grade crossings** (outside business/residence districts); super-heavy equipment (crawler tractors, steam shovels, derricks, rollers) and explosives trucks must also stop.
- Passing motorcycles/bicycles: allow **at least 6 ft** to their left and do not share the lane — move into the next lane to pass.

Registration

- Out-of-state plates are recognized for **30 days** after you establish residency or take gainful employment.
- Buyer's temporary cardboard tags are valid for **20 days**. Vehicles are registered in the county of residence.

2 · General Knowledge (Driving Safely + Cargo)

Vehicle inspection

- The **7-step inspection method**: (1) vehicle overview, (2) engine compartment, (3) start engine & check the cab, (4) lights, (5) walk-around, (6) signal lights, (7) brake-system check.
- Out-of-adjustment brakes are the **most common** defect found in roadside inspections.

Speed & stopping distance (at 55 mph, ideal conditions)

- Total stopping distance = **perception + reaction + braking ≈ 419 ft** at 55 mph. Perception ≈ 1.5 sec (≈142 ft); reaction ≈ ¾-1 sec (≈61 ft); braking ≈ 216 ft (dry, good brakes).
- **Empty trucks need a greater stopping distance** — they have less traction; brakes/tires are designed for the loaded weight.
- Wet roads can **double** stopping distance. Reduce speed about ⅓ on wet roads (55→~35 mph), about ½ on packed snow, and crawl on ice.
- Speed math: doubling speed (20→40 mph) makes braking distance **4x**; tripling (20→60) makes it **9x**. At 60 mph your stopping distance is longer than a football field.
- Hydroplaning can start as low as **30 mph**; more likely with low tire pressure or worn tread. If hydroplaning, release the accelerator and don't brake.
- Bridges freeze before roads — be careful near **32°F**; watch for black ice and shaded spots.
- Curves: a high-center-of-gravity truck can roll over **at the posted curve speed** — slow down **before** the curve, not in it.

Space & communicating

- Following distance: at least **1 second for every 10 ft of vehicle length** below 40 mph; add **1 more second above 40 mph** (a 40-ft vehicle at 55 mph = 5 seconds).
- Signal early and keep a clear escape path; always have a plan for hazards.

Cargo

- Inspect cargo within the **first 50 miles**, then re-check after every **3 hours or 150 miles**.
- Securement: at least **one tie-down per 10 ft of cargo**, and a minimum of **two tie-downs** no matter how short the load.
- Know the weight terms: **GVWR** (single vehicle), **GCWR** (combination), axle weight, tire load. Keep the load **centered and low** to prevent rollover; max gross 80,000 lb.
- Use blocking and bracing to stop cargo from sliding; header boards ('headache racks') protect you from forward-shifting cargo.

3 · General Knowledge — Weather, Hazards & Emergencies

Night, fog & weather

- Night: clean your headlights, mirrors, windshield and glasses; **don't overdrive your headlights** — you must be able to stop within the distance you can see (low beams light only ~250 ft). Dim high beams within **500 ft** of oncoming traffic; use high beams when safe and legal; never stare at oncoming lights.
- Fog: **avoid driving** in fog if you can; use **low beams / fog lights** (high beams reflect back and worsen glare); slow down and watch for stopped vehicles.
- Winter: check coolant/antifreeze, defroster, wipers/washer fluid, tires, lights and the exhaust system (carbon-monoxide leaks); make sure the parking brake won't freeze; bridges ice over first.
- Hot weather: heat raises tire pressure — check it; watch engine coolant and oil; bleeding tar makes pavement slippery on very hot days; if a tire gets too hot, stop and let it cool.

Mountain / downgrade driving

- Choose a safe speed for the **weight, length and steepness** of the grade and the weather; never exceed a posted 'Maximum Safe Speed'.

- **Engine braking is the main control** — be in the proper **low gear before** the grade; you can't safely downshift once speed builds.
- Snub braking: brake to ~5 mph below your safe speed, release for ~3 sec, then repeat.
- Know the **escape-ramp (runaway-truck ramp)** locations on your route and use one if your brakes fail.

Skids & emergencies

- Most common skid = rear drive-wheel skid (too much braking or acceleration). Acceleration skid → get off the gas (push the clutch if very slippery).
- Braking-skid recovery: **stop braking** to let the wheels roll, then **counter-steer** (and quickly steer back the other way as it straightens).
- Front-wheel skid: usually too much speed — the only fix is to **slow down** (stop turning/braking so hard).
- Stopping isn't always safest — **steering away** can be quicker; if you must leave the road, going **right** is usually best.
- Brake failure: **downshift**, pump the brakes, use the parking/emergency brake, and find an escape route (uphill or an escape ramp).
- Tire blowout: hold the wheel **firmly with both hands**, stay **off the brake** until the vehicle is under control, then brake gently and pull off. Signs: loud bang, vibration, or steering pull.
- ABS: **don't pump it** — press and hold. ABS keeps wheels from locking and helps you steer; it doesn't necessarily shorten stopping distance. If the ABS lamp stays on, you still have normal brakes.

Fires

- Fire classes: **A** = wood/paper/cloth (water or dry chemical); **B** = gasoline/oil/grease (CO2 or dry chemical — **not water**); **C** = electrical (CO2 or dry chemical — **never water**).
- Pull off into an **open area**, away from buildings and other vehicles; **do not pull into a service station**; notify emergency services.
- Engine fire: shut off the engine and don't open the hood if you can avoid it. Cargo fire in a van/box: **keep the doors shut** (opening adds oxygen).
- Aim the extinguisher at the **base** of the fire, stay as far back as possible and upwind.

Alcohol, drugs & staying fit

- CDL limit: you are legally **intoxicated at .04% BAC** or more (lower than the .08% car limit); driving a CMV with **any detectable alcohol** is a violation.
- Only **time** sobers you up — coffee, fresh air and exercise don't. More alcohol and faster drinking = higher BAC.
- Don't drive while ill, fatigued, or on drugs (including some prescription/OTC meds) that impair you. **Drowsy driving** is a major crash cause — get enough sleep.

Accident procedures

- Three steps: **protect the area, notify authorities, care for the injured**.
- Don't move a seriously injured person unless there's danger (e.g., fire); keep the injured warm and stop heavy bleeding.

4 · General Knowledge — Vehicle Control, Turning & Space

Basic control (accelerate, steer, stop)

- Accelerate **smoothly and gradually** so the vehicle doesn't jerk — rough acceleration can damage the coupling when pulling a trailer. On poor traction (rain/snow) speed up very gradually; if the drive wheels spin, ease off the accelerator.

- Don't **roll back** when starting on a hill: partly engage the clutch before releasing the brake, or hold with the parking brake (or the trailer hand valve on a tractor-trailer).
- Steer with **both hands on opposite sides** of the wheel and a firm grip — a curb or pothole can jerk the wheel away.
- Stop **smoothly**: apply the brake gradually; with a manual transmission, push the clutch in only when the engine is near idle.

Shifting gears

- Know when to **shift up** using engine speed (tachometer / operating rpm range) or road speed (speedometer); many newer trucks use **progressive shifting**.
- **Double-clutch** if your vehicle requires it; **downshift before** going down a hill or into a curve, at the right rpm.
- Coasting in neutral or with the clutch in is unlawful and unsafe — stay in gear.

Looking ahead & mirrors

- Look far enough ahead — at least **12-15 seconds** (the distance you'll travel in that time).
- Check mirrors **regularly**, and more often before lane changes and turns; watch for vehicles entering your **blind spots**; adjust mirrors before the trip.

Signaling

- Signal **early** and signal **continuously** — don't cancel the turn signal until the turn is finished (and turn it off afterward if it doesn't self-cancel).
- Put your signal on **before** changing lanes, and change lanes slowly and smoothly.
- Warn drivers behind you when slowing — **tap the brake pedal** to flash the lights; use hazards if you're going very slow or stopped.

Turning (right turns)

- Turn **slowly** to give yourself and others time.
- If you can't turn right without swinging into another lane, **turn wide as you complete** the turn — keep the rear near the curb so others can't pass on your right.
- **Don't swing wide to the left** before turning right (a follower may think you're going left and try to pass on the right).
- Texas method: approach the corner ~4 ft from the curb; as the front wheels pass the corner, turn so the **right rear wheel clears the curb**.
- If you must cross into the oncoming lane to turn, give oncoming vehicles room — but **don't back up** for them (you could hit someone behind).

Space around the rig

- Overhead: **don't trust posted bridge heights** (re-paving or snow can reduce them); an **empty van rides higher** than a loaded one; check before backing under branches/wires.
- Below: watch low clearance, drainage dips, and railroad crossings — don't get **hung up**.
- Crossing/entering traffic: slow acceleration means you need a **much bigger gap** than a car; allow more room when heavily loaded; make sure you can get **all the way across** before traffic reaches you.

5 · General Knowledge — Inspection, Equipment, Railroad & Hazards

Tires & wheels (what to check)

- Tread depth: at least **4/32 inch** in every major groove on **front (steering) tires**, and **2/32 inch** on all other tires.
- No **regrooved, recapped or retreaded** tires on the **front wheels of a bus**.

- Defects to catch: mismatched tire sizes, cut/cracked valve stems, cuts or bulges, tread separation, cracked/damaged wheels or rims, loose or missing lug nuts, and low pressure.

Required emergency equipment

- A **properly charged and rated fire extinguisher**.
- **Spare electrical fuses** — unless the vehicle has circuit breakers.
- Warning devices: **3 red reflective triangles, OR 6 fusees, OR 3 liquid-burning flares**. Set them out to warn traffic whenever you stop on the road.

Fire extinguisher type

- A **B:C** extinguisher works on **electrical fires and burning liquids**; an **A:B:C** also works on wood/paper/cloth.
- **Don't use water** on an electrical or gasoline fire; water may be used on wood/paper/cloth or to cool a burning tire.

Railroad crossings (driving)

- **Never race a train**; don't expect to hear it; don't rely only on signals or gates.
- **Never get trapped** where you'd have to stop on the tracks; don't start across unless you can get all the way over.
- **Do not shift gears while crossing** the tracks.
- A typical tractor-trailer needs **at least 14 seconds to clear a single track** and **more than 15 seconds for a double track**.
- (Section 14) Buses carrying passengers for hire, school buses, and explosives trucks must stop at all crossings.

Hazardous materials awareness

- Vehicles carrying hazmat in amounts that require **placards** must display them; you need the **HazMat (H) / combination (X) endorsement** to haul placardable hazmat.
- **Recognize placards** — they identify the hazard class. Knowing them is basic CDL knowledge even without the endorsement.

Seeing hazards & staying fit

- A **hazard** is any road condition or road user that is a possible danger; spotting it early gives you time to check mirrors, signal, slow down, or change lanes.
- Watch for **drivers who are hazards**: blocked vision, slow/parked/delivery vehicles, distracted or confused drivers, and **impaired or drowsy drivers** (especially late at night).
- **Fatigue** impairs you like alcohol — get enough sleep, take breaks, and don't drive when too tired.

6 · General Knowledge — Inspection Details, Cargo & Communicating

Inspection — key defects to catch

- Steering: more than **10 degrees of steering-wheel play** (about 2 in. on a 20-in. wheel) makes it hard to steer; check the column, gear box and tie rods; on power steering check hoses, pump, fluid and leaks.
- Suspension: no broken, shifted or missing leaf springs or parts that let an axle move out of place.
- Wheels/rims: **no welding repairs**; rust streaks around the lug nuts can mean **loose nuts** — check tightness.
- Brakes: shoes or pads with **oil, grease or brake fluid** are bad; watch for thin or cracked drums.
- Tires: **don't mix radial and bias-ply** tires, and dual tires must not touch each other (plus tread depth 4/32 front, 2/32 others).

Inspection — engine & in-cab

- Engine compartment: check engine oil, coolant, power-steering fluid, belts and hoses, and look for leaks.
- In the cab: confirm the parking brake holds; test the **horn, wipers/washer, lights, heater/defroster**; check mirrors and windshield for cracks/illegal stickers; watch the gauges (oil pressure, coolant/oil temp, air pressure, voltmeter) and warning lights; **wear your seat belt**.

Cargo — weight & balance

- **You are responsible** for not being overloaded and for a properly balanced load.
- Overloading hurts **steering, braking and control**; too much weight or a **high center of gravity** makes the rig harder to stop and easier to roll over.
- Keep weights within legal limits (GVWR / GCWR / axle limits; max **80,000 lb** gross).

Communicating your presence

- Turn on your **headlights** (not just marker lights) when it's hard to see — at dawn, dusk, or in rain.
- Parked at the roadside: turn on the **4-way emergency flashers**, especially at night — drivers do crash into parked trucks.
- Use your **horn** only when needed; tap it lightly (or flash your lights at night) to signal.

Distracted driving

- It is **illegal for CMV drivers to text or use a hand-held phone** while driving — it brings fines and can lead to **disqualification**.
- Reaching for, dialing, or holding a phone to talk all count as banned hand-held use; employers may not require it.

7 · CDL Classes, Rules & Disqualifications

CDL classes

- **Class A**: any combination with a GCWR of **26,001 lb or more** where the towed unit(s) exceed **10,000 lb** GVWR — i.e. tractor-trailers (your class).
- **Class B**: a single vehicle of **26,001 lb+** GVWR (may tow a unit $\leq 10,000$ lb), or any vehicle built to carry **24+ passengers** including the driver.
- **Class C**: a vehicle that isn't Class A or B but carries **16-23 passengers** (incl. driver) or **placardable hazardous materials**.

Endorsements & restrictions

- Endorsements add privileges: **P** = passenger, **S** = school bus, **N** = tank vehicle, **T** = doubles/triples, **H** = hazardous materials, **X** = hazmat + tank combined.
- A **restriction** limits you — e.g. an **air-brake restriction** if you don't pass the air-brakes test or test in a vehicle without air brakes, or an **automatic-transmission restriction** if you test in an automatic.
- A **Commercial Learner Permit (CLP)** lets you practice under a CDL holder before you take the skills test.

Alcohol & major offenses

- It is illegal to operate a CMV at **.04% BAC** or more; by driving you give **implied consent** to alcohol testing — **refusing the test** is itself a disqualifying offense.
- You lose your CDL for **at least 1 year (first offense)** for any of: driving a CMV at .04%+ BAC, DUI of alcohol or a controlled substance, refusing a test, **leaving the scene** of a CMV accident, using a CMV to commit a **felony**, driving a CMV with a suspended CDL, or causing a **fatality** by negligent driving.
- **At least 3 years** if the offense happens while the CMV is **placarded for hazmat**.

- **Lifetime** for a second major offense; lifetime if you use a CMV for a felony involving controlled substances.
- Put **out of service for 24 hours** if you have **any detectable alcohol** under .04%.

Serious traffic violations

- Examples: excessive speeding (**15+ mph over**), reckless driving, improper/erratic lane changes, following too closely, fatal-accident traffic offenses, and driving without your CDL / proper class / endorsements.
- Lose your CDL **≥60 days** for 2 within 3 years; **≥120 days** for 3 or more within 3 years.

Out-of-service order violations

- **≥90 days** for the first; **≥1 year** for 2 within 10 years; **≥3 years** for 3 or more within 10 years.

Railroad-crossing violations

- **≥60 days** for the first; **≥120 days** for a second within 3 years; **≥1 year** for a third within 3 years.

8 · Air Brakes

How the system works

- Air brakes are really **three systems**: service, parking, and emergency.
- Air compressor **governor**: cut-out **≈ 125 psi** (stops pumping), cut-in **≈ 100 psi** (resumes).
- **Safety relief valve** opens at **150 psi**; if it releases air, something is wrong — see a mechanic.
- **Low air pressure warning** must come on **before pressure falls below 55 psi** (this Texas handbook). On large buses it often signals at 80–85 psi.
- **Spring (parking/emergency) brakes** apply automatically when pressure drops to **20–45 psi** (typically 20–30) — don't wait for this; stop safely as soon as the warning comes on.
- Dual air system: before driving, build at least **100 psi in both** systems; the warning should turn off above 55 psi.
- **Brake lag**: air takes $\sim\frac{1}{2}$ second to act, so total stopping = perception + reaction + brake lag + braking. At 55 mph brake lag adds ~ 32 ft → total over **450 ft**.
- **ABS** required: tractors built on/after **March 1, 1997**; trucks, buses, trailers, dollies on/after **March 1, 1998**. Yellow malfunction lamp. ABS doesn't shorten stopping distance — it helps you keep control and steer.

Inspecting air brakes (Step 7 checks)

- Slack adjusters: if one moves **more than ~1 inch** where the push rod attaches, it needs adjustment.
- Brake drums must not have cracks longer than **half the friction-area width**; linings must not be loose, oil-soaked, or dangerously thin.
- Air-leak test (engine off, brakes **released**): loss should be **under 2 psi/min (single)** and **under 3 psi/min (combination)**.
- Air-leak test with brakes **applied** (hold 1 min): **under 3 psi (single) / 4 psi (combination)**.
- Pressure build-up: **85→100 psi within 45 seconds** in dual systems.
- Also test: the low-pressure warning, that spring brakes apply automatically, the parking brake (tug in low gear), and the service brakes at ~ 5 mph.

Using air brakes

- Emergency stops: use **controlled braking** (as hard as possible without locking) or **stab braking** (full apply, release when wheels lock, reapply — wheels can take up to 1 sec to roll again).
- Downgrades: braking is only a **supplement to engine braking**; be in the proper low gear before you descend. Snub technique: brake to ~ 5 mph below your safe speed, release ~ 3 sec, repeat.
- If the low-air warning comes on while driving, **stop and park as soon as possible**.

- Don't set the parking brakes if they are very hot (just came down a grade) or very wet in freezing temps — they can be damaged or freeze. Use wheel chocks; never leave a vehicle without the parking brake or chocks.

9 · Combination Vehicles

Driving combinations

- Rollover: more than half of truck-driver deaths are rollovers; a **fully loaded** rig is **10x more likely** to roll than an empty one. Two keys: keep cargo **low** and drive **slowly** around turns; keep the load centered.
- **Crack-the-whip** (rearward amplification): a quick lane change can flip the trailer; the rear trailer of triples is **3.5x** more likely to roll than the tractor. Steer gently and smoothly.
- Brake early: empty and **bobtail** (tractor without trailer) combinations take **longer** to stop than a loaded tractor-trailer.
- **Off-tracking** ('cheating'): the rear wheels follow a **tighter/shorter** path than the front; longer rigs off-track more. **Turn wide** — steer the front out enough so the rear clears the curb, but keep the rear near the curb so others can't pass on the right.
- Backing a trailer: turn the steering wheel the **opposite** way at first, then follow the trailer; back toward the **driver's side**, slowly, with a helper.
- Railroad crossings: low-clearance trailers (lowboy, car carrier, moving van, possum-belly) can get stuck; if stuck, get out and away from the tracks and call 911 with the DOT crossing number.
- Trailer skid: spot it early in your mirrors; **release the brakes** to regain traction; **do not** use the trailer hand valve to straighten the rig.

Combination air brakes

- **Trailer hand valve** (trolley / Johnson bar): use it **only to test** the trailer brakes — never while driving (skid risk) and **never for parking** (air can leak off).
- **Tractor protection valve**: closes automatically at **20-45 psi** to keep air in the tractor if the trailer breaks away.
- **Trailer air supply control**: a red **8-sided** knob; push in to supply air, pull out for the trailer emergency brakes; it pops out at 20-45 psi.
- Two air lines: **service** (control/signal) and **emergency** (supply). Losing emergency-line pressure applies the trailer emergency brakes.
- **Glad hands** (couplers): connect at a **90° angle**; **blue = service, red = emergency**. Crossed lines mean no trailer brakes — very dangerous; always test before driving.
- Shut-off (cut-out) valves: open on every trailer **except the back of the last trailer**, which is closed.
- Trailers/dollies built **before 1975** aren't required to have spring brakes → use **wheel chocks** when parking them.

Coupling & uncoupling (key checks)

- The fifth-wheel locking jaws must close around the **shank (base) of the kingpin** — not the top. The kingpin must not be bent or broken.
- Fifth wheel tilted **down toward the rear** of the tractor; back slowly until it just touches the trailer; the kingpin must be aligned and centered.
- **Tug test**: raise the landing gear slightly, lock the trailer brakes, and pull gently forward in low gear to confirm the lock holds.
- Connect air lines and the electrical cord, secure everything, then **visually inspect** the coupling — there must be **no gap** (daylight) between the fifth wheel and the trailer.
- Walk-around (combination): fifth wheel **securely mounted & greased**, all **locking pins** in place, no visible space between upper/lower fifth wheel, air/electric lines connected with **slack for turns** and no leaks, and the **landing gear fully raised** with the crank handle secured.

Study aid summarized from the Texas Commercial Driver License Handbook. Always confirm against the current official handbook and DPS before testing.